

Ayush Kashyap — Shipping AI Into a Workflow People Actually Trust

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THE PROBLEM

Doctors lose hours a day to documentation. The obvious fix — record the consultation, transcribe it, have an LLM write the note — is a demo in a week and a failed product in a month. I built the version that survived contact with real clinics.

WHAT I BUILT

- **Ambient AI medical scribe** in production: audio capture → streaming speech-to-text → LLM-generated structured clinical notes → clinician review and sign-off, inside a multi-tenant EHR serving **2,000+ clinics** and hundreds of thousands of patients.
- **Speech pipeline**: speech-to-text for real-world Indian clinic audio (noise, code-switched Hindi-English, overlapping speakers), including speech-model fine-tuning to close the gap generic models left.
- **Retrieval grounding**: prior patient context pulled in to anchor generation rather than letting the model invent history.

THE THREE DECISIONS THAT MADE IT WORK

1. Optimise for “fast to correct”, not “never wrong.” Nothing persists until a clinician signs. That single constraint converts an unsolvable accuracy problem into a solvable editing-speed problem — and it is why the product was safe to ship.

2. Treat fields by blast radius, not uniformly. A hallucinated adjective in prose is noise; a hallucinated dosage or a flipped negation (“no history of diabetes”) is harm. High-risk fields get stricter extraction, validation and surfacing than narrative text.

3. Evaluate on production diffs, not synthetic sets. The signal that mattered was what clinicians actually changed between generated and signed notes. That became the eval loop — and it caught failure modes no curated test set contained.

LATENCY IS THE PRODUCT

Adoption did not turn on model quality alone. If transcription lags the conversation, the tool stops being ambient and becomes a distraction. Much of the real engineering was streaming architecture, latency budgets, graceful degradation and fallbacks when a model call failed mid-consultation — the unglamorous layer that decides whether an AI feature gets used twice.

HOW I WORK WITH AI DAY TO DAY

Agentic coding tools are core workflow, not novelty: Claude Code, MCP servers and custom agent harnesses for scaffolding, refactors, migrations and review — with architecture, correctness and the final judgement call staying mine. The same principle as the product: let AI compress idea → shipped, keep a human on anything where being wrong has a cost.

STACK

TypeScript · JavaScript · Python · React · React Native · Node.js / Express · MySQL · Redis · Socket.io · AWS · Docker/K8s · FHIR R4 / HL7